Question

1. What is a conditional statement in programming?

Ans –

A conditional statement is used to make decisions in a program by executing different blocks of code based on whether a condition is true or false.

Question

- 2. What does an if statement do?

Ans –

An if statement checks a condition. If the condition is true, it executes the block of code inside the if statement.

Question

- 3. What is the purpose of an else block?

Ans –

The else block executes a block of code when the condition in the if statement is false.

Question

- 4. When is an else block executed?

Ans-

An else block is executed only when the condition in the corresponding if statement evaluates to false.

Question

- 5. What is a Boolean expression?

Ans –

A Boolean expression is an expression that evaluates to either true or false.

Question

- 6. Write the syntax of a simple if statement.

Ans –

```
if (condition)
{
 // statements
 }
```

Question

- 7. What operator is commonly used to test equality?

Ans –

The equality operator == is used to check whether two values are equal.

Question

- 8. What does the modulus (%) operator return?

Ans –

The modulus (%) operator returns the remainder after dividing one number by another.

Question

- 9. How can you check if a number is even?

Ans-

- In programming, the most common and efficient way to check if a number is even is by using the modulus operator (%), which returns the remainder of a division.
- Since any even number is perfectly divisible by 2, dividing it by 2 leaves a remainder of 0.

Ex-

```
if (number % 2 == 0) {
  // The number is even
}
```

Question

- 10. How can you check if a number is odd?

Ans –

Since odd numbers always leave a remainder of 1 (or a non-zero value for negative numbers) when divided by 2, you check if the remainder is not equal to 0

Ex-

```
if (number % 2 != 0) {
   // The number is odd
}
```

Question

- 11. What is the purpose of else-if?

Ans –

The else-if statement is used to test multiple conditions one after another. If the first if condition is false, the program checks the next else-if condition. The block corresponding to the first true condition is executed.

Question

- 12. What is a nested if statement?

Ans –

A nested if statement is an if statement placed inside another if or else block. It is used when a second condition should be checked only if the first condition is satisfied.

Question

- 13. Can an if statement exist without an else?

Ans-

An if statement can be used without an else block. In this case, the code inside the if block executes only when the condition is true. If the condition is false, the program simply skips that block.

Question

- 14. What is the output when a false condition is tested in an if statement without else?

Ans –

When the condition is false and there is no else block, no code inside the if statement is executed. The program continues with the next statement after the if block.

Question

- 15. What is the role of braces {} in C/C++ conditionals?

Ans –

Braces {} are used to group multiple statements into a single block. They define the code that belongs to an if, else, or other conditional statement. Although optional for a single statement, using braces improves readability and helps prevent programming errors.

Question

- 16. What does > mean?

Ans-

The > operator is the greater than relational operator. It checks whether the value on the left is greater than the value on the right. It returns true if the condition is satisfied; otherwise, it returns false.

Question

- 17. What does < mean?

Ans-

The < operator is the less than relational operator. It checks whether the value on the left is smaller than the value on the right.

Question

- 18. What does >= mean?

Ans-

The >= operator means greater than or equal to. It returns true if the left value is either greater than or exactly equal to the right value.

Question

- 19. What does <= mean?

Ans-

The <= operator means less than or equal to. It returns true if the left value is either less than or equal to the right value.

Question

- 20. What does != mean?

Ans-

The != operator is the not equal to operator. It compares two values and returns true if they are different; otherwise, it returns false.

Question

- 21. What is a switch statement?

Ans-

A switch statement is a decision-making control structure that allows a program to execute one block of code from multiple possible options based on the value of an expression. It is commonly used as an alternative to long if-else-if ladders when comparing constant values.

Question

- 22. What is a case label in a switch statement?

Ans –

A case label represents a specific constant value in a switch statement. If the value of the switch expression matches a case label, the corresponding block of code is executed.

Question

- 23. Why is break used in switch?

Ans-

The break statement is used to terminate the execution of a switch case. It prevents the program from executing the statements in the following cases.

Question

- 24. What happens if break is omitted?

Ans-

If the break statement is omitted, the program continues executing the next case(s) even if their values do not match. This behavior is known as fall-through.

Question

- 25. What is the default case in switch?

Ans-

The default case contains the code that is executed when none of the case labels match the value of the switch expression. It is optional but recommended for handling unexpected values.

Question

- 26. Can switch compare strings in C++?

Ans-

No. In standard C++, a switch statement cannot directly compare strings. It is mainly used with integral types such as int, char, and enum. String comparisons are usually performed using if-else statements.

Question

- 27. Which statement is better for many constant choices: if-else or switch?

Ans-

A switch statement is generally better when comparing a single variable against many constant values because it is more readable, easier to maintain, and often more efficient than a long if-else-if ladder.

Question

- 28. How do you test whether a value is positive?

Ans-

A value is positive if it is greater than zero.

if (num > 0)

Question

- 29. How do you test whether a value is negative?

Ans-

A value is positive if it is greater than zero.

if (num > 0)

Question

- 30. How do you test whether a value is zero?

Ans-

A value is zero if it is equal to zero.

```
if (num == 0)
```

Question

- 31. Write a condition to check if age is at least 18.

Ans-

To check whether a person is eligible for voting or other age-based criteria, the age should be greater than or equal to 18.

```
if (age >= 18)
```

Question

- 32. Write a condition to check if marks are above 50.

Ans-

To verify whether a student has scored more than 50 marks, use the following condition:

if (marks > 50)

Question

- 33. What is decision making in programming?

Ans-

Decision making is the process of selecting one course of action from multiple alternatives based on whether a condition is true or false. It allows programs to respond differently to different inputs using conditional statements such as if, if-else, and switch.

Question

- 34. Can nested if statements have multiple levels?

Ans-

Yes. A nested if statement can have multiple levels, meaning one if statement can be placed inside another repeatedly. This is useful for solving complex decision-making problems that require checking several conditions in sequence.

Question

- 35. What is logical AND (&&)?

Ans-

The logical AND (&&) operator combines two or more conditions. It returns true only if **all** the conditions are true. If any one condition is false, the entire expression evaluates to false.

Example:

- if (age >= 18 && age <= 60)

Question

- 36. What is logical OR (||)?

Ans-

- The logical OR (||) operator is used to combine multiple conditions. It returns true if at least one of the conditions is true. It returns false only when all conditions are false.

Example:

```
if (marks >= 90 || grade == 'A')
```

Question

- 37. What is logical NOT (!)?

Ans-

- The logical NOT (!) operator reverses the result of a Boolean expression. If the condition is true, it becomes false; if it is false, it becomes true.

Example:

if (!isLoggedIn)

Question

- 38. How do you check if a year is divisible by 4?

Ans-

A year is divisible by 4 if the remainder after dividing it by 4 is zero.

Example-

```
if (year % 4 == 0)
```

Question

- 39. What is a leap year?

Ans-

A leap year is a year that contains **366 days** instead of 365. According to the Gregorian calendar, a leap year is divisible by 4, except for century years, which must also be divisible by 400.

Question

- 40. What is the first condition for a leap year?

Ans-

The first condition for checking a leap year is that the year must be divisible by **4**.

```
if (year % 4 == 0)
```

Note: A complete leap year check also verifies whether century years are divisible by **400**.

Question

- 41. How do you find the largest of two numbers?

Ans-

- The largest of two numbers can be found by comparing them using the greater-than (>) operator. If the first number is greater than the second, it is the largest; otherwise, the second number is the largest.

Example:

```
if (a > b)
   cout << "a is the largest";
else
   cout << "b is the largest";
```

Question

- 42. How do you find the largest of three numbers?

Ans-

- To find the largest of three numbers, compare each number using an if-else-if ladder. The number that is greater than or equal to the other two is the largest.

Example:

```
if (a >= b && a >= c)
   cout << "a is the largest";
else if (b >= a && b >= c)
   cout << "b is the largest";
else
   cout << "c is the largest";
```

Question

- 43. What is an if-else-if ladder?

Ans-

An if-else-if ladder is a sequence of multiple conditions checked one after another. The program executes the block of the **first condition that evaluates to true**. If none of the conditions are true, the else block (if present) is executed.

Question

- 44. What is the difference between if and switch?

Ans-

if Statement	switch Statement
Can evaluate complex conditions using relational and logical operators.	Compares a single expression with constant values only.
Suitable for ranges and multiple logical conditions.	Best for selecting one option from many fixed choices.

Question

- 45. Can switch use ranges directly?

Ans-

No. A switch statement cannot directly evaluate ranges such as 1–10 or 50–100. It can only compare an expression with constant values. For checking ranges, an if-else statement should be used.

Question

- 46. What is fall-through in switch?

Ans-

Fall-through is the behavior in which the program continues executing the statements of the next case(s) after a matched case if the break statement is omitted. This continues until a break is encountered or the switch statement ends.

Question

- 47. What data type is usually used for conditions?

Ans-

The bool (Boolean) data type is commonly used for conditions. It stores either true or false, which determine whether a block of code should be executed.

Question

- 48. Why are conditional statements important?

Ans-

Conditional statements are important because they enable a program to make decisions based on different situations. They improve the flexibility, efficiency, and intelligence of a program by allowing it to execute different actions depending on the input or conditions

Question

- 49. Give one real-life example of a conditional statement.

Ans-

A common real-life example is an **ATM machine**. If the entered PIN is correct, the user is allowed to access their account; otherwise, access is denied.

Question

- 50. Name four conditional structures covered.

Ans-

The four conditional structures covered are:

1.if statement

2.if-else statement

3.if-else-if ladder

4.switch statement